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REVIEW ARTICLE

Comparative Efficacy of Continuous Positive Airway Pressure and Antihypertensive Medications in Obstructive Sleep Apnea-Related Hypertension: A Narrative Review

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Abstract

Introduction Obstructive sleep apnea (OSA) presents a significant global health concern, affecting a substantial portion of the population, particularly among young and middle-aged adults.

Aim This review aims to assess the efficacy of continuous positive airway pressure (CPAP) compared to antihypertensive medications in managing OSA-related hypertension.

Methods A comprehensive literature search was conducted across multiple databases, yielding studies published from 2000 to March 2024 that investigated CPAP, antihypertensives, or their combination therapy in OSA patients. Six definitive studies were analyzed, including two randomized controlled trials (RCTs), one randomized double-blind placebo-controlled crossover trial, one placebo-controlled trial, one open-label multicenter trial, and one longitudinal cohort study. These studies comprised 939 participants, with intervention durations ranging from four weeks to six months.

Results Analysis of CPAP monotherapy revealed variable efficacy, with some studies demonstrating significant reductions in 24-hour mean blood pressure and diastolic pressure, while others reported non-significant changes. CPAP therapy combined with antihypertensives showed additive effects, particularly in reducing office blood pressure measurements. Antihypertensive medications, such as valsartan, exhibited superior efficacy in reducing blood pressure compared to CPAP alone. Factors influencing therapy effectiveness included CPAP compliance, patient characteristics, and coexisting comorbidities. Patients with good CPAP adherence experienced greater reductions in blood pressure. The duration of exposure to OSA and the type of hypertension also impacted the therapy response.

Conclusion While CPAP and antihypertensive medications offer significant benefits in managing hypertension among OSA patients, challenges such as CPAP intolerance and medication side effects exist. Personalized treatment considering individual patient factors is crucial for optimal management.

Keywords Hypertension · Obstructive sleep apnea · CPAP · Anti-hypertensives · Obesity

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1 Introduction

Obstructive sleep apnea (OSA), characterized by frequent breathing cessation during sleep [1], is a prevalent sleeping disorder affecting a substantial portion of the global population, particularly among young and middle-aged adults [2]. Recent estimates indicate that nearly one billion adults aged thirty to sixty-nine years have mild to severe OSA, with 425 million falling into the moderate to severe category [2, 3]. The rise in OSA prevalence correlates with the increasing obesity rates observed in developed countries. While variations exist across populations and age groups, studies consistently report OSA's impact on approximately 24% to 26% of men and 17% to 28% of women between the ages of 30 and 70 [4, 5]. Notably, males exhibit a higher prevalence, which escalates with age.

OSA's etiology involves multifactorial mechanisms, with recognized risk factors including obesity, aging, gender, large neck size, and craniofacial abnormalities. Additionally, OSA often presents with comorbidities spanning metabolic, cardiovascular, renal, pulmonary, and neuropsychiatric domains [5]. Among these, hypertension emerges as the most prevalent complication, significantly affecting morbidity and mortality rates [4, 5].

Untreated hypertension in obstructive sleep apnea (OSA) patients can precipitate severe complications, including heart failure due to pulmonary hypertension and an increased risk of coronary artery disease, recurrent heart attacks, and stroke [6–8]. Furthermore, untreated hypertension heightens the likelihood of various arrhythmias, exacerbating cardiovascular adverse events, morbidity, and mortality [9–11]. Recognized as a cause of secondary hypertension in a significant proportion of cases, OSA warrants screening and treatment, particularly in patients exhibiting symptoms such as interrupted sleep and snoring, alongside those with resistant hypertension [12]. Treatment modalities encompass both non-pharmacological and pharmacological approaches [13–15].

Non-pharmacological interventions, including continuous positive airway pressure (CPAP), mandibular advancement devices (MAD), sleep surgery, and weight loss strategies, are pivotal in OSA management [12, 16]. CPAP, recommended as the first-line treatment, effectively maintains upper airway patency during sleep and significantly reduces OSA severity and blood pressure, especially in cases of resistant hypertension [13, 14, 17]. While CPAP demonstrates efficacy, its effectiveness in controlling blood pressure may vary based on factors such as baseline blood pressure levels and patient compliance [18–20]. Similarly, MADs offer a viable alternative for CPAP-intolerant individuals, showing comparable effectiveness in blood pressure reduction [21, 22]. Additionally, sleep surgery has shown promise in reducing blood pressure in adults with OSA [23–25].

Pharmacological interventions remain crucial for achieving optimal blood pressure control in hypertensive patients with OSA [26, 27]. Understanding the underlying mechanisms of OSA-related hypertension informs the selection of antihypertensive medications, with agents targeting sympathetic and renin-angiotensin-aldosterone systems showing potential efficacy [28–30]. Comparing the efficacy of CPAP therapy to that of antihypertensive medication is essential for informing evidence-based practices in hypertension management. This review aims to synthesize findings from diverse clinical trials assessing the efficacy of CPAP and antihypertensive drugs in managing OSA-related hypertension.

Table 1 Methodology

Methodology Step	Description
1.	Thorough search of literature in databases such as PubMed, Google Scholar, COCHRANE, EMBASE, and SCOPUS
2.	Use of keywords (“CPAP,” “Obstructive Sleep Apnea,” “hypertension,” “blood pressure,” “antihypertensives”) and Medical Subject Headings (MeSH) terms for search refinement
3.	Inclusion criteria: Studies published from 2000, including clinical trials and cohort studies investigating CPAP and antihypertensives, as well as those comparing monotherapy with combination therapy
4.	Exclusion criteria: Narrative reviews, cross-sectional studies, and studies evaluating single therapy efficacy without a comparative element
5.	Language criterion: Only studies available in full text and published in English were considered
6.	Data extraction by two independent authors (IJO & AEB) including details such as author and year, study design, sample size, intervention, duration, outcome measures, and results

2 Methods

The writing of this review involved a thorough search of existing literature. Several databases, such as PubMed, Google Scholar, COCHRANE, EMBASE, and SCOPUS, were searched to find articles aligned with the study's objectives. Using keywords like “CPAP,” “Obstructive Sleep Apnea,” “hypertension,” “blood pressure,” and “antihypertensives,” these databases were meticulously combed through, employing both specific keywords and Medical Subject Headings (MeSH) terms to refine our search (Table 1).

Studies published from 2000 till March 2024 pertinent to our research aims were included. This encompassed clinical trials, cohort studies investigating both CPAP and

antihypertensives and studies comparing CPAP or antihypertensive monotherapy with combination therapy. Only studies available in full text and published in English were considered. Narrative reviews and cross-sectional studies were excluded. Studies evaluating the efficacy of a single therapy without a comparative element, be it CPAP or antihypertensives, were also excluded.

Two independent authors (IJO & AEB) extracted data from the included studies, recording details such as author and year, study design, sample size, intervention, duration, outcome measures, and results. The authors collaborated to reach conclusions regarding the extracted data to ensure thoroughness and precision. A central illustration of the summary of the main findings and conclusions is shown in Fig. 1

3 Results

A thorough literature review informed our selection of vital studies examining the comparative efficacy of CPAP and antihypertensive medications in controlling patients' blood pressure with a diagnosis of OSA—Table 2. Thus, six definitive studies have been selected, including two randomized controlled trials (RCTs), one randomized double-blind placebo-controlled crossover trial, one placebo-controlled trial, one open-label multicenter trial, and one longitudinal cohort study. Collectively, the patient population amounts to 939 participants with a duration of intervention ranging from four weeks to six months. The explored interventions include the different modalities of application of CPAP and a range of antihypertensive medications such as losartan, perindopril, and valsartan, thus ensuring unique perspectives on outcomes of therapeutic choices across different clinical settings.

4 Efficacy of CPAP Alone

In the randomized placebo-controlled trial fundamental study by Campos-Rodriguez et al. [31], both the therapeutic and subtherapeutic (placebo) CPAP were administered to patients with hypertension and OSA. A small, statistically non-significant decrease in 24-hour mean blood pressure was noted after four weeks of therapeutic CPAP administration [31]. Interestingly, Martinez-Garcia et al.'s [32] study in 2013 on patients with resistant hypertension, including the category of patients who were not responsive to traditional antihypertensive medications, revealed that CPAP treatment resulted in a distinctive decrease in 24-hour mean blood pressure and diastolic blood pressure over 12 weeks. In addition, a positive correlation between hours of CPAP use

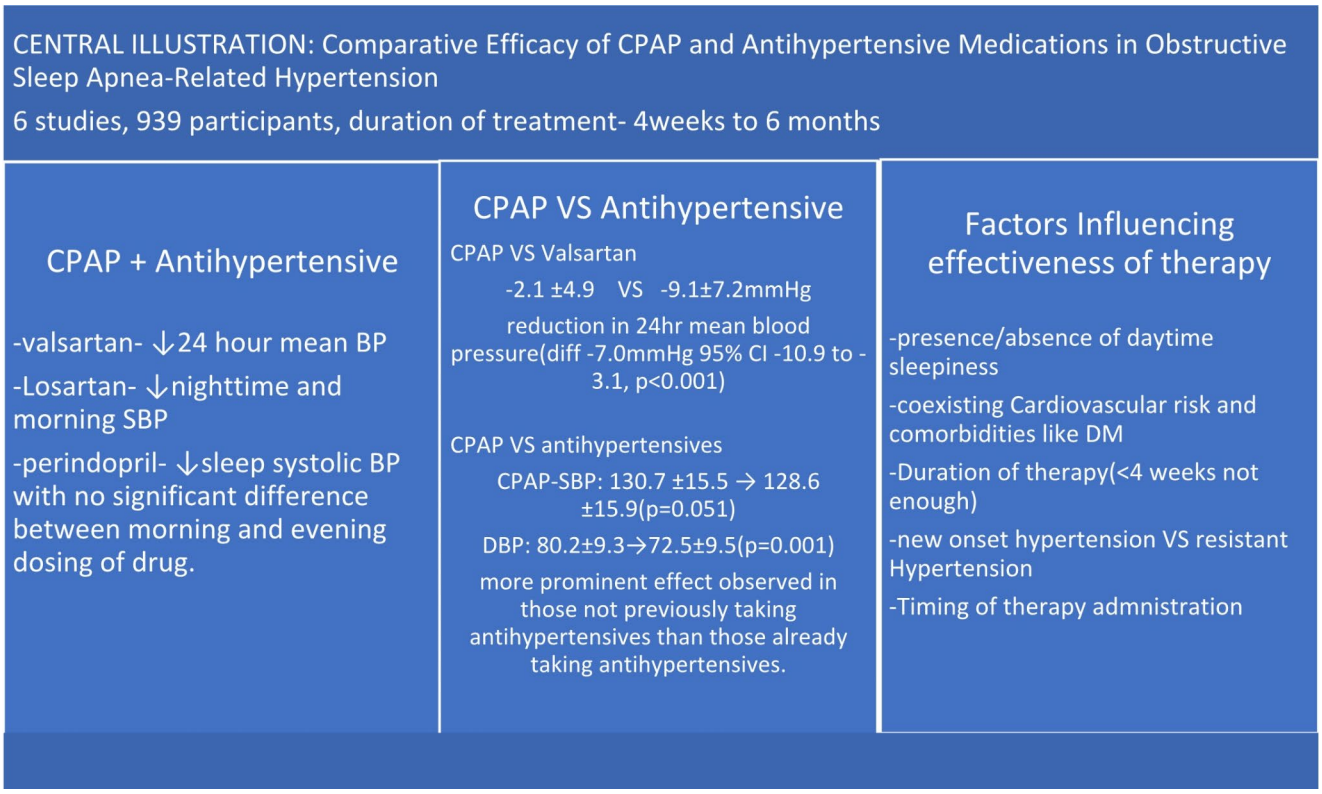

Fig. 1 Central Illustration

Table 2 Characteristics of Included Studies

References	Study design	Sample Size	Intervention (CPAP vs. Antihypertensives)	Duration of Intervention	Outcome Measures	Results
Pépin et al. [34]	Randomized controlled cross-over trial	23	CPAP (a treatment for OSA) against the antihypertensive drug valsartan (160 mg)	8 weeks, followed by a 4-week wash-out period to minimize carryover effects between treatments.	The primary outcome measure was the change in 24-hour mean blood pressure (BP) assessed by ambulatory BP monitoring (ABPM) at the end of each treatment period. Secondary outcome measures included changes in daytime and nighttime BP, as well as other clinical parameters such as heart rate and biochemical parameters.	The study found that while CPAP treatment led to a modest reduction in 24-hour mean BP (−2.1 ± 4.9 mm Hg), valsartan was significantly more effective, resulting in a greater reduction (−9.1 ± 7.2 mm Hg, P < 0.001). The difference in 24-hour mean BP reduction between valsartan and CPAP was statistically significant (−7.0 mm Hg; 95% CI, −10.9 to −3.1 mm Hg; P < 0.001). Similar trends were observed for daytime and nighttime BP reductions. The study also noted that the combination of CPAP and valsartan had additive effects on BP reduction, particularly for office BP measurements, although the effect was less pronounced for 24-hour BP.
Thunström et al. [35]	*Phase 1*: Open, prospective, parallel-design, case-control study. *Phase 2*: Sex-stratified, open, randomized parallel-design study within the OSA group.	*Initially* 89 patients were included 54 with OSA, 35 without. *Final analysis* after exclusion of some partcipants included 24 OSA patients who received CPAP plus losartan, 23 OSA patients who received only losartan, and 34 patients without OSA who continued on losartan alone	Losartan 50mg daily to all patients. CPAP was added as an intervention in Phase 2 for OSA patients.	12 weeks (6 weeks for each phase)	Primary End-points: 24-hour systolic blood pressure (SBP), diastolic blood pressure (DBP), and mean arterial pressure (MAP). Secondary End-points: Daytime, nighttime, and morning BP measurements, evaluation of dipping versus non-dipping patterns, and proportion of optimally treated patients.	Phase 1: Losartan significantly reduced SBP, DBP, and MAP in both OSA and non-OSA groups. Patients with OSA showed slightly less BP reduction compared to those without OSA, particularly during morning hours. Only 13.0% of OSA patients achieved optimal BP levels after 6 weeks of losartan treatment. Phase 2: In the intention-to-treat (ITT) analysis, CPAP as an add-on to losartan resulted in nonsignificant changes in primary 24-hour BP values. However, significant reductions were observed in nighttime and morning SBP with CPAP. Adherent CPAP use (≥4 hours/night) showed significant reductions in all 24-hour BP values. Optimal BP levels increased significantly in the CPAP group, particularly among adherent patients.

Table 2 (continued)

References	Study design	Sample Size	Intervention (CPAP vs. Antihypertensives)	Duration of Intervention	Outcome Measures	Results
Campos-Rodriguez et al. [31]	Parallel, randomized, placebo-controlled trial.	68	Therapeutic CPAP or subtherapeutic CPAP (considered as the placebo) Antihypertensive treatment remained unchanged throughout the study period.	4 weeks	The primary outcome measure was the change in ambulatory blood pressure (BP) after treatment. Additional outcome measures included changes in systolic and diastolic BP, daytime and nighttime BP, and restoration of the normal circadian dipper pattern.	Both therapeutic and subtherapeutic CPAP groups showed a small, statistically nonsignificant decrease in 24-hour mean BP after 4 weeks of treatment. There were no significant changes observed in systolic, diastolic, daytime, or nighttime BP. The restoration of the normal circadian dipper pattern was slightly higher in the therapeutic CPAP group, though not statistically significant. Importantly, there was no correlation found between the change in 24-hour mean BP and CPAP compliance, severity of OSAS, or the number of antihypertensive drugs used.
Martínez-García et al. [32]	open-label, randomized, multicenter clinical trial	194	Patients in the CPAP group received CPAP treatment for 12 weeks. The control group received no CPAP treatment but continued their usual antihypertensive medication regimen.	12 weeks	Primary Endpoint: Change in 24-hour mean blood pressure after 12 weeks. Secondary Endpoints: Changes in other blood pressure values (systolic and diastolic), and changes in nocturnal blood pressure patterns (e.g., dipper pattern).	The study found that CPAP treatment led to a greater decrease in 24-hour mean blood pressure (3.1 mm Hg) and 24-hour diastolic blood pressure (3.2 mm Hg) compared to the control group. The CPAP group also exhibited an increased percentage of patients with a nocturnal blood pressure dipper pattern compared to the control group (35.9% vs 21.6%; adjusted odds ratio [OR], 2.4). There was a significant positive correlation between hours of CPAP use and the decrease in blood pressure levels (mean, systolic, and diastolic).

Table 2 (continued)

References	Study design	Sample Size	Intervention (CPAP vs. Antihypertensives)	Duration of Intervention	Outcome Measures	Results
Serinel et al. [36]	Randomized, double-blind, placebo-controlled crossover trial in 3-phases. Phase III was an open-label phase	85 (78 patients completed the entire trial)	Evening versus morning administration of perindopril with placebo. In phase III, CPAP therapy was added to the treatment regimen for all patients.	6 weeks (phases I and II), followed by an 8-week open-label phase with CPAP (phase III)	The primary outcome measure was the difference in sleep systolic blood pressure (SBP) determined by 24-hour ambulatory blood pressure monitoring (ABPM) between morning and evening ingestion of perindopril. Secondary outcomes included sleep diastolic blood pressure (DBP), wake and 24-hour SBP and DBP, BP dipping ratios, and office BP.	The study found that both morning and evening dosing of perindopril resulted in significant reductions in sleep and wake SBP and DBP compared to baseline. There was no significant difference in sleep SBP reduction between morning and evening dosing (1.1 mm Hg more reduction with morning dosing, but not statistically significant). Wake SBP was reduced significantly more with morning dosing compared to evening dosing (1.8 mm Hg). CPAP therapy further reduced sleep SBP (morning dosing: -3.3 mm Hg, evening dosing: -3.2 mm Hg) but did not show a difference in improvement between morning and evening dosing. Non-dipping rates (a marker of cardiovascular risk) did not improve significantly with perindopril dosing or with the addition of CPAP therapy.
Börgel et al. [33]	Longitudinal cohort study	540	Bilevel or continuous positive airway pressure (Bi-/CPAP) therapy for OSA. This intervention was compared with the effects of antihypertensive medications, which were either maintained or adjusted during the study period.	6 months	The primary outcome measures included changes in daytime systolic and diastolic BP, as well as HR after 6 months of Bi-/CPAP therapy. These changes were evaluated in relation to the severity of OSA (measured by AHI) and the presence or absence of concurrent antihypertensive medication.	The study found significant improvements in daytime systolic BP (from 130.7 ± 15.5 to 128.6 ± 15.9 mm Hg, P = 0.051), diastolic BP (from 80.2 ± 9.3 to 77.5 ± 9.5 mm Hg, P = 0.001), and HR (from 77.7 ± 8.8 to 75.7 ± 8.1 beats/min, P = 0.001) after 6 months of effective Bi-/CPAP therapy. Multiple linear regression analysis identified that the absence of antihypertensive drugs and higher initial BP values were significant predictors of the lowering effect of Bi-/CPAP therapy on systolic and diastolic BP. Notably, the reduction in BP was more pronounced in patients not taking antihypertensive medications. In contrast, patients already on antihypertensive treatment showed a lesser response to Bi-/CPAP therapy in terms of BP reduction, although HR reduction was observed irrespective of antihypertensive medication use.

and the reduction in blood pressure levels brings to the fore a possible temporal factor in the effect of CPAP in controlling blood pressure. It establishes that CPAP therapy alone may reduce blood pressure over a protracted administration period and potentially challenging scenarios. Additionally, the longitudinal cohort study by Borgel et al. [33] in 2004 further highlights the benefits of CPAP therapy administration alone as it revealed significant improvements in the daytime systolic and diastolic blood pressures in patients using Bi-/CPAP therapy, especially in the participants who weren't currently being managed with any antihypertensive medication, over six months [33]. This also reinforces the potential efficacy of CPAP therapy in reducing blood pressure over a relatively long period.

Cross-examination of these studies shows that CPAP has a variable efficacy across different contexts and scenarios. It may emanate from several factors, including the severity of sleep apnea, patients' compliance with therapy, and the physiological response of each participant across the studies. However, there seems to be a temporal relationship between CPAP duration of administration and its effectiveness in decreasing the 24-hour mean blood pressure and CPAP's potential to be a valuable non-pharmacological intervention for patients with resistant hypertension and/or those not wanting long-term pharmacological therapy is established.

4.1 CPAP Combination with Antihypertensives

The randomized controlled cross-over trial study by Pepin et al. [34], which compared the individual and combined effects of Continuous Positive Airway Pressure (CPAP) and the angiotensin-2 receptor-blocking antihypertensive agent, Valsartan showed that there was a reduction in the 24-hour mean blood pressure on the administration of both treatments indicating an additive effect and particularly noticeable in the office blood pressure measurements [34]. In another two-phase cohort study with OSA patients where CPAP was added to a losartan treatment, while isolated CPAP didn't significantly alter 24-hour blood pressure metrics in the intention-to-treat analysis, however, remarkable reductions were noted in the nighttime and morning systolic blood pressure among patients compliant with CPAP usage (defined as ≥4 hours/night). This emphasizes the significance of compliance to therapy and indicates that CPAP can augment the efficacy of ongoing antihypertensive therapy, especially during nighttime [35]. Both studies provide insights into the benefits of synergistic efforts to reduce the mean blood pressure using CPAP and antihypertensive medications. Interestingly, Serinel et al. [36] randomized double-blind placebo-controlled crossover trial study [36] exploring the impact of perindopril dosing times and the addition of CPAP revealed that the addition of CPAP resulted in a further reduction in the sleep systolic blood pressure and more importantly, there was no significant difference in the effect between morning and evening dosing hence, establishing that in terms of the therapy efficacy, there is no difference between the dosing times.

4.2 CPAP Versus Antihypertensives

In the Pepin et al. [34] study, the effects of CPAP and valsartan, an angiotensin-2 receptor-blocking antihypertensive agent, to reduce the 24-hour mean blood pressure assessed by Ambulatory Blood pressure Monitoring (ABPM) were directly compared with each other. Valsartan proved more effective than CPAP as it reduced -9.1 ± 7.2 mm Hg compared to -2.1 ± 4.9 mm Hg with CPAP. This statistically significant difference in efficacy establishes pharmacological therapy (Valsartan) as a superior option for controlling blood pressure over CPAP alone [34]. Martinez-Garcia et al. [32] also examined the impact of CPAP as compared to the traditional form of care, which included antihypertensive medications in a category of patients with characteristic resistant hypertension. The study revealed that CPAP significantly reduced 24-hour mean and diastolic blood pressure compared to the control group [32]. This indicates that while CPAP is less effective than valsartan, as depicted by Pepin et al. [34], it significantly reduces patients with resistant hypertension, thus potentially improving sleep quality and oxygen saturation during sleep.

Borgel et al. [33] then explore the long-term effects of CPAP therapy compared to the antihypertensive medications effect on the blood pressure in the participants. It was found that CPAP therapy resulted in a marginal reduction and a marginal statistical non-significance (from 130.7 ± 15.5 to 128.6 ± 15.9 mm Hg, $P = 0.051$) in the systolic blood pressure and a significant reduction and statistical significance in the diastolic pressure (from 80.2 ± 9.3 to 77.5 ± 9.5 mm Hg, $P = 0.001$) indicating that CPAP was effective as a primary intervention in patients with mild hypertension and Obstructive Sleep Apnea. Statistical analysis further revealed that this effect was more pronounced in patients not taking any antihypertensive medication, but a lesser response was observed in patients taking antihypertensive medications.

4.3 Factors Influencing the Effectiveness of Therapies

Multiple studies have evaluated the factors influencing the effectiveness of therapies for obstructive sleep apnea (OSA). A study by Pépin et al. [34] suggests that the efficacy of CPAP treatment can vary depending on factors such

as the presence or absence of daytime sleepiness [37], co-existing cardiovascular risk factors such as diabetes mellitus, and whether the patient already has cardiovascular problems. Moreover, personalized treatment for OSA is crucial, considering metabolic and cardiovascular risks, patient compliance, and lifestyle modifications. Additionally, the combination of CPAP and valsartan resulted in a significant decrease in blood pressure [38, 39], particularly at night, for patients who did not achieve adequate control with either therapy alone. This suggests that the combination of therapies and timing of treatment administration may influence its effectiveness.

Another study by Thunström et al. [35] suggested that the duration of exposure to OSA and the type of hypertension (new-onset vs. therapy-resistant) may affect the response to different antihypertensive treatments [40]. A Placebo-Controlled Trial conducted by Campos-Rodriguez et al. [31] indicated that CPAP treatment may need longer to significantly lower blood pressure in patients with OSA and hypertension treated with antihypertensives [40]. A 4-week treatment period was insufficient to demonstrate significant blood pressure changes. Furthermore, co-morbid conditions such as obesity and type II diabetes mellitus can exacerbate hypertension [41], potentially limiting the effectiveness of CPAP treatment on blood pressure.

Furthermore, Martínez-García et al. [32] found a notable association between the number of hours of CPAP use and the decrease in blood pressure levels. Patients with good adherence to CPAP treatment, especially those using CPAP for at least 4 hours per night, tend to experience greater reductions in blood pressure [42]. In their 2017 study, Serinel et al. [36] highlighted the role of considering circadian variation [43] and CPAP compliance in optimizing therapy efficacy for managing hypertension in patients with OSA. Börgel et al.'s longitudinal cohort study [33] showed that CPAP therapy had a greater impact on blood pressure in patients with higher initial blood pressure levels and those not using antihypertensive medication. This indicates that CPAP therapy's efficacy may differ based on individual patient characteristics like baseline blood pressure and medication usage [44].

5 Discussion

Analyzing the efficacy of CPAP monotherapy revealed a significant decrease in 24-hour mean blood pressure and diastolic pressure and a positive correlation between the hours of CPAP use and the reduction in blood pressure in two studies [32, 33]. Another study, however, concluded that the blood pressure reduction observed in their study was statically insignificant. One study that analyzed dual therapy of CPAP and antihypertensives reported that both therapies might benefit patients with hypertension and OSA due to their synergistic action [31]. However, this study also noted that valsartan, an angiotensin two receptor blocker, was more effective than CPAP in reducing 24-hour mean blood pressure [31]. This finding was, however, inconsistent with that of another study, which noted that CPAP was more effective in the long-term reduction of 24-hour mean blood pressure and diastolic pressure [32]. Pepin et al. [34] suggested that the efficacy of CPAP treatment depended on factors including daytime sleepiness and existing comorbidities.

The findings also revealed a few significant implications for clinical practice in managing hypertension among patients with OSA. For instance, the severity of sleep apnea, existing comorbidities, and medical tolerance were identified as key in treatment selection and personalized care of patients [34]. CPAP may be beneficial as the primary intervention for patients with mild hypertension and OSA, especially those who are not currently on antihypertensives [33]. It may also be effective as an adjunct for patients with resistant hypertension who are not achieving adequate blood pressure control with traditional antihypertensives [32]. This is also true for patients with moderate and severe hypertension with OSA [33, 34]. While CPAP therapy and antihypertensive medications offer significant benefits in managing hypertension among patients with OSA, a few drawbacks and challenges were noted with their use in clinical practice. Regarding monotherapy with CPAP, issues including compliance with CPAP therapy owing to discomfort, claustrophobia, or adjustment difficulties associated with its use, as well as lack of generalization of results associated with its use, makes its applicability difficult [44]. Similarly, side effects associated with the use of antihypertensives may pose difficulty with their use as monotherapy for patients [44]. Additionally, the cost and complexity of using both interventions may hamper their use as dual therapy for patients [44].

The few potential sources of bias are identifiable in the studies, potentially affecting the results obtained. For instance, the presence of confounding variables, including age, gender, sex, body mass index, smoking status, existing comorbidities, and chronic drug use, may influence treatment outcomes and introduce bias in the results. Additionally, variations in study designs, including differences in intervention protocols, outcome measures, and follow-up durations, may introduce methodological limitations. Moving on, there remains a need to address other key areas of the subject, including comparative long-term cardiovascular outcomes (such as risk of stroke, myocardial infarction, and heart failure) associated with both interventions used as mono or dual therapy. Additionally, future research may

benefit from comparing both interventions with other novel approaches, including pharmacological agents and other treatment modalities.

6 Limitations and Strengths of Review

While this review provides valuable insights into the efficacy of CPAP and antihypertensive medications in managing hypertension associated with OSA, it is important to acknowledge limitations. Firstly, there may be a risk of publication bias as only studies published in English and available in full text were included, potentially excluding relevant non-English publications or unpublished studies. The variability in study designs, patient populations, intervention protocols, and outcome measures among the included studies may limit direct comparisons and the generalizability of findings. Despite these limitations, the review possesses strengths. The review provides a detailed analysis of the efficacy of CPAP and antihypertensive medications, including monotherapy and combination therapy, offering insights into their comparative effectiveness in managing OSA-related hypertension. Furthermore, the review discusses the implications of the findings for clinical practice, highlighting the potential benefits and challenges associated with different treatment modalities, thereby informing evidence-based decision-making in hypertension management among OSA patients.

7 Conclusions

This review study shows the complex interplay between CPAP and antihypertensives for blood pressure control in obstructive sleep apnea. Although there has been a moderate reduction in 24-hour mean blood pressure with CPAP, an ARB Antihypertensive losartan significantly decreased 24-hour mean blood pressure, systolic, diastolic, and mean arterial blood pressure in obstructive sleep apnea. This only insinuates that pharmacological management is paramount in effectively controlling blood pressure. Furthermore, combining CPAP and Antihypertensive resulted in a non-significant change in 24-hour blood pressure. However, there was an additive effect on office blood pressure measurements. Notably, the reduction in blood pressure with CPAP when a patient was already on antihypertensives had a lesser response compared to when they were not on antihypertensives.

OSA and hypertension often occur together; even though CPAP causes a modest decrease in blood pressure, its use should not be discouraged as it is considered in the management guidelines as an add-on treatment for hypertension in obstructive sleep apnea. Management decisions should be based on the individual patient's degree of hypertension and severity of obstructive sleep apnea, the presence of other comorbidities, and the patient's tolerance to medications. Patients with both disease conditions typically start with CPAP therapy, as it improves sleep apnea and controls blood pressure, even though modestly. Antihypertensives may also be added if elevated blood pressure persists despite CPAP therapy. As stated earlier, the management decisions should be individualized depending on the patient's response to therapy and condition. It is common knowledge that OSA and hypertension often coexist with other diseases like diabetes and other cardiovascular diseases. Therefore, to achieve a good overall health outcome, these comorbidities must be comprehensively managed to prevent treatment resistance.

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Authors' Contributions ODA conceptualized the study; all authors were involved in the literature review; IJO & AEB extracted the data from the reviewed studies; All authors wrote the final and first drafts. All authors read and approved the final manuscript.

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Availability of Data and Material Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

Code Availability Not applicable.

Declarations

Competing interests The authors declare that they have no competing interests.

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

References

1. Surani S, Taweesedt P. Obstructive Sleep Apnea: New Perspective. Medicina (Kaunas). 2022;59(1):75.
2. Arredondo E, Udeani G, Panahi L, Taweesedt PT, Surani S. Obstructive Sleep Apnea in Adults: What Primary Care Physicians Need to Know. Cureus. 2021;13(9):e17843.
3. Benjafield AV, Ayas NT, Eastwood PR, Heinzer R, Ip MSM, Morrell MJ, Nunez CM, Patel SR, Penzel T, Pépin JL, Peppard PE, Sinha S, Tufik S, Valentine K, Malhotra A. Estimation of the global prevalence and burden of obstructive sleep apnoea: a literature-based analysis. Lancet Respir Med. 2019;7(8):687–98. https://doi.org/10.1016/S2213-2600(19)30198-5.
4. Cunningham J, Hunter M, Budgeon C, Murray K, Knuiman M, Hui J, Hillman D, Singh B, James A. The prevalence and comorbidities of obstructive sleep apnea in middle-aged men and

women: the Busselton Healthy Ageing Study. J Clin Sleep Med. 2021;17(10):2029–39. https://doi.org/10.5664/jcsm.9378.
5. Fietze I, Laharnar N, Obst A, Ewert R, Felix SB, Garcia C, Gläser S, Glos M, Schmidt CO, Stubbe B, Völzke H, Zimmermann S, Penzel T. Prevalence and association analysis of obstructive sleep apnea with gender and age differences—Results of SHIP-Trend. J Sleep Res. 2019;28(5):e12770. https://doi.org/10.1111/jsr.12770.
6. Wang Y, Li CX, Lin YN, Zhang LY, Li SQ, Zhang L, Yan YR, Lu FY, Li N, Li QY. The Role of Aldosterone in OSA and OSA-Related Hypertension. Front Endocrinol (Lausanne). 2022;12(12): 801689. https://doi.org/10.3389/fendo.2021.801689.PMID:35095768;PMCID:PMC8791261.
7. Cai A, Wang L, Zhou Y. Hypertension and obstructive sleep apnea. Hypertens Res. 2016;39(6):391–5.
8. Ahmad M, Makati D, Akbar S. Review of and Updates on Hypertension in Obstructive Sleep Apnea. Int J Hypertens. 2017;2017:1–13.
9. Gunta SP, Jakulla RS, Ubaid A, Mohamed K, Bhat A, López-Candales A, Norgard N. Obstructive Sleep Apnea and Cardiovascular Diseases: Sad Realities and Untold Truths regarding Care of Patients in 2022. Cardiovasc Ther. 2022;11(2022):6006127. https://doi.org/10.1155/2022/6006127.
10. Wahab A, Chowdhury A, Jain NK, Surani S, Mushtaq H, Khedr A, Mir M, Jama AB, Rauf I, Jain S, Korsapati AR, Chandramouli MS, Boike S, Attallah N, Hassan E, Chand M, Bawaadam HS, Khan SA. Cardiovascular Complications of Obstructive Sleep Apnea in the Intensive Care Unit and Beyond. Medicina (Kaunas). 2022;58(10):1390. https://doi.org/10.3390/medicina58101390.
11. Parsons C, Mookadam F, Mookadam M. The effect of obstructive sleep apnea treatment on cardiovascular outcomes. Future Cardiol. 2017;13(1):5–8.
12. Jordan AS, McSharry DG, Malhotra A. Adult obstructive sleep apnoea. Lancet. 2014;383:736–47.
13. Kushida CA, Littner MR, Hirshkowitz M, Morgenthaler TI, Aleessi CA, Bailey D, Boehlecke B, Brown TM, Coleman J Jr, Friedman L, Kapen S, Kapur VK, Kramer M, Lee-Chiong T, Owens J, Pancer JP, Swick TJ, Wise MS. Practice parameters for the use of continuous and bilevel positive airway pressure devices to treat adult patients with sleep-related breathing disorders. Sleep. 2006;29(3):375–80.
14. Gottlieb DJ, Punjabi NM, Mehra R, Patel SR, Quan SF, Babineau DC, Tracy RP, Rueschman M, Blumenthal RS, Lewis EF, Bhatt DL, Redline S. CPAP versus oxygen in obstructive sleep apnea. N Engl J Med. 2014;370(24):2276–85.
15. Durán-Cantolla J, Aizpuru F, Montserrat JM, Ballester E, Terán-Santos J, Aguirregomoscorta JI, Gonzalez M, Lloberes P, Masa JF, De La Peña M, Carrizo S, Mayos M, Barbé F. Continuous positive airway pressure as treatment for systemic hypertension in people with obstructive sleep apnoea: randomized controlled trial. BMJ. 2010;341:c5991.
16. Bratton DJ, Gaisl T, Wons AM, Kohler M. CPAP vs mandibular advancement devices and blood pressure in patients with obstructive sleep apnea: a systematic review and meta-analysis. JAMA. 2015;314(21):2280–93.
17. Posadas T, Francisco Campos-Rodriguez F, Sapiña-Beltrán E, Oscullo G, Torres G, Martinez-Garcia MA. Obstructive sleep apnea and arterial hypertension: implications of treatment adherence. Curr Hypertens Rep. 2020;22(2):12.
18. Drager LF, Polotsky VY, Lorenzi-Filho G. Obstructive sleep apnea: an emerging risk factor for atherosclerosis. Chest. 2011;140(2):534–42.
19. Rotenberg BW, Murariu D, Pang KP. Trends in CPAP adherence over twenty years of data collection: a flattened curve. J Otolaryngol Head Neck Surg. 2016;45(1):43.
20. Baratta F, Pastori D, Bucci T, Fabiani M, Fabiani V, Brunori M, Loffredo L, Lillo R, Pannitteri G, Angelico F, Ben MD. Long-term prediction of adherence to continuous positive air pressure therapy for the treatment of moderate/severe obstructive sleep apnea syndrome. Sleep Med. 2018;43:66–70.
21 de Vrie GE, Wijkstra PJ, Houwerzijl EJ, Kerstjens HAM, Hoekema A. Cardiovascular effects of oral appliance therapy in obstructive sleep apnea: a systematic review and meta-analysis. Sleep Med Rev. 2018;40:55–68.
22. Pengo MF, Soranna D, Giontella A, Perger E, Mattaliano P, Schwarz EI, Lombardi C, Bilo G, Zambon A, Steier J, Parati G, Minuz P, Fava C. Obstructive sleep apnoea treatment and blood pressure: which phenotypes predict a response? A systematic review and meta-analysis. Eur Respir J. 2020;55(5):1901945.
23. Schiavon CA, Bhatt DL, Ikeoka D, Santucci EV, Santos RN, Damiani LP, Oliveira JD, Machado RHV, Halpern H, Monteiro FLJ, Noujaim PM, Cohen RV, de Souza GG, Amodeo C, Bortolotto LA, Berwanger O, Cavalcanti AB, Drager LF. Three-year outcomes of bariatric surgery in patients with obesity and hypertension: a randomized clinical trial. Ann Intern Med. 2020;173(9):685–93.
24. Kang KT, Yeh T, Ko J, Lee C, Hsu W. Effect of sleep surgery on blood pressure in adults with obstructive sleep apnea: a systematic review and meta-analysis. Sleep Med Rev. 2022;62.
25. Kent D, Stanley J, Aurora RN, Levine C, Gottlieb DJ, Spann MD, Torre CA, Green K, Harrod CG. Referral of adults with obstructive sleep apnea for surgical consultation: an American academy of sleep medicine systematic review, meta-analysis, and GRADE assessment. J Clin Sleep Med. 2021;17(12):2507–31.
26. Jiang W, Li W, Cheng J, Li W, Cheng F. Efficacy and safety of liraglutide in patients with type 2 diabetes mellitus and severe obstructive sleep apnea. Sleep Breath. 2023;27(5):1687–94.
27. Chirinos JA, Gurubhagavatula I, Teff K, Rader DJ, Wadden TA, Townsend R, Foster GD, Pack AI. CPAP, weight loss, or both for obstructive sleep apnea. N Engl J Med. 2014;370(24):2265–75.
28 Gaddam K, Pimenta E, Thomas SJ, Cofield SS, Oparil S, Harding SM, Calhoun DA. Spironolactone reduces severity of obstructive sleep apnoea in patients with resistant hypertension: a preliminary report. J Hum Hypertens. 2010;24(8):532–7. https://doi.org/10.1038/jhh.2009.96.
29. Ziegler MG, Milic M, Elayan H. Cardiovascular regulation in obstructive sleep apnea. Drug Discov Today Dis Model. 2011;8(4):155–60.
30. Kraiczi H, Hedner J, Peker Y, Grote L. Comparison of atenolol, amlodipine, enalapril, hydrochlorothiazide, and losartan for antihypertensive treatment in patients with obstructive sleep apnea. Am J Respir Crit Care Med. 2000;161(5):1423–8.
31. Campos-Rodriguez F, Grilo-Reina A, Perez-Ronchel J, Merino-Sanchez M, Gonzalez-Benitez MA, Beltran-Robles M, Almeida-Gonzalez C. Effect of continuous positive airway pressure on ambulatory BP in patients with sleep apnea and hypertension: a placebo-controlled trial. Chest. 2006;129(6):1459–67. https://doi.org/10.1378/chest.129.6.1459.
32. Martínez-García MA, Capote F, Campos-Rodríguez F, Lloberes P, Díaz de Atauri MJ, Somoza M, Masa JF, González M, Sacristán L, Barbé F, Durán-Cantolla J, Aizpuru F, Mañas E, Barreiro B, Mosteiro M, Cebrián JJ, de la Peña M, García-Río F, Maimó A, Zapater J, Hernández C, Grau SanMarti N, Montserrat JM; Spanish Sleep Network. Effect of CPAP on blood pressure in patients with obstructive sleep apnea and resistant hypertension: the HIPARCO randomized clinical trial. JAMA. 2013;310(22):2407–15. https://doi.org/10.1001/jama.2013.281250.
33. Börgel J, Sanner BM, Keskin F, Bittlinsky A, Bartels NK, Büchner N, Huesing A, Rump LC, Mügge A. Obstructive sleep apnea and blood pressure. Interaction between the blood pressure-lowering effects of positive airway pressure therapy and antihypertensive drugs. Am J Hypertens. 2004;17(12 pt 1):1081–7. https://doi.org/10.1016/j.amjhyper.2004.06.026.

34. Pépin JL, Tamisier R, Barone-Rochette G, Launois SH, Lévy P, Baguet JP. Comparison of continuous positive airway pressure and valsartan in hypertensive patients with sleep apnea. Am J Respir Crit Care Med. 2010;182(7):954–60. https://doi.org/10.1164/rccm.200912-1803OC.
35. Thunström E, Manhem K, Rosengren A, Peker Y. Blood pressure response to losartan and continuous positive airway pressure in hypertension and obstructive sleep apnea. Am J Respir Crit Care Med. 2016;193(3):310–20. https://doi.org/10.1164/rccm.201505-0998OC.
36. Serinel Y, Yee BJ, Grunstein RR, Wong KH, Cistulli PA, Arima H, Phillips CL. Chronotherapy for hypertension in obstructive sleep apnoea (CHOSA): a randomised, double-blind, placebo-controlled crossover trial. Thorax. 2017;72(6):550–8. https://doi.org/10.1136/thoraxjnl-2016-209504.
37. Robinson GV, Smith DM, Langford BA, Davies RJ, Stradling JR. Continuous positive airway pressure does not reduce blood pressure in nonsleepy hypertensive OSA patients. Eur Respir J. 2006;27(6):1229–35.
38. Hu X, Fan J, Chen S, Yin Y, Zrenner B. The role of continuous positive airway pressure in blood pressure control for patients with obstructive sleep apnea and hypertension: a meta-analysis of randomized controlled trials. The Journal of Clinical Hypertension. 2015;17(3):215–22.
39. Xu SD, Hao LL, Liu FF, Xu CZ. The effects of obstructive sleep apnea on blood pressure variability and load in patients with hypertension. Sleep and Breathing. 2024;7:1.
40. Kapa S, SertKuniyoshi FH, Somers VK. Sleep apnea and hypertension: interactions and implications for management. Hypertension. 2008;51(3):605–8.
41. Modan M, Almog S, Fuchs Z, Chetrit A, Lusky A, Halkin H. Obesity, glucose intolerance, hyperinsulinemia, and response to antihypertensive drugs. Hypertension. 1991;17(4):565–73.
42. Sánchez-De-La-Torre M, Gracia-Lavedan E, Benitez ID, Sánchez-De-La-Torre A, Moncusí-Moix A, Torres G, Loffler K, Woodman R, Adams R, Labarca G, Dreyse J. Adherence to CPAP treatment and the risk of recurrent cardiovascular events: a meta-analysis. JAMA. 2023;330(13):1255–65.
43. Hermida RC, Ayala DE, Smolensky MH, Fernández JR, Mojón A, Crespo JJ, Ríos MT, Moyá A, Portaluppi F. Chronotherapeutics of conventional blood pressure-lowering medications: simple, low-cost means of improving management and treatment outcomes of hypertensive-related disorders. Curr Hypertens Rep. 2014;16(2):412.
44. Pengo M, Schwarz EI, Barbe F, Drager L, Fava C, Ip MS, Martinez Garcia MA, McEvoy D, Peker Y, Phillips CL, Soranna D. Effect of CPAP therapy on blood pressure in patients with OSA: a worldwide individual patient data meta-analysis. Eur Heart J. 2022;43(Supplement_2):ehac544-2453.

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